

CSA0704 — COMPUTER NETWORKS

Department of Computer Science and Engineering

Design and Performance Evaluation of a Resilient Smart Campus Network

using Modern Layered Protocols

Team Assignment Report

Role	Team Member	Register No.	Primary Responsibility
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Course Outcomes Addressed: CO1–CO5

SDG Mapping: SDG 4 · SDG 9 · SDG 11

Maximum Marks: 100

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1. Problem Understanding & Formulation

The university operates four principal buildings across a multi-campus footprint — an Administrative Block, an Academic Block, a Hostel/IoT zone, and a central Data Centre — that must be interconnected into a single resilient network. The existing infrastructure suffers from five concrete failure modes: wireless collisions during high-density lecture and exam periods, inefficient flat IP allocation that wastes address space and complicates access control, slow routing-table convergence after link failures, TCP congestion collapse during peak hours when thousands of students stream lecture content simultaneously, and unreliable application services such as delayed DNS resolution, dropped email delivery, and file transfers with no confidentiality guarantee.

1.1 Entities Identified

- End users: ~5,000 concurrent wireless clients (students, faculty, staff, guests) and 500 IoT energy-monitoring sensors.
- Network devices: Layer-3 core/distribution routers, Layer-2 access switches, Wi-Fi 6 (802.11ax) access points, an SDN controller for the Data Centre segment, and a firewall/NAT edge router.
- Services: DNS resolver hierarchy, HTTP/HTTPS web/LMS portal with TLS 1.3, SMTP/IMAP mail, secure file transfer for research data, and SNMP-based monitoring.
- Traffic classes: bulk research data transfer (up to 1 Gbps between buildings), interactive lecture streaming, IoT telemetry (small, periodic), and administrative/transactional traffic.

1.2 Assumptions

- Simulation is carried out in Cisco Packet Tracer with Wireshark used for protocol-level capture of representative traffic; where Packet Tracer cannot generate genuine congestion/queuing behaviour, results are supplemented with analytical queuing-theory estimates.
- Each building has independent power and a dedicated uplink to the core, allowing an active/standby backbone design.
- A single top-level domain, campus.edu, is used for internal DNS with a split-horizon design for internal vs external resolution.

1.3 Constraints Carried Into the Design

Constraint	Target	Design Response
Concurrent wireless users	$\geq 5,000$	802.11ax with OFDMA + hierarchical /20–/22 wireless VLANs
IoT devices	≥ 500	Dedicated IoT VLAN (BLE gateway + Wi-Fi) sized to /23
Wireless packet loss	$< 1\%$	CSMA/CA + OFDMA scheduling, TWT, RF-aware AP placement
DNS resolution (local)	< 50 ms	Local authoritative + caching resolver per building, low TTL tuning
TCP goodput recovery	≤ 2 RTT after loss	CUBIC congestion control + DiffServ QoS marking
Routing convergence	< 5 s (single link failure)	OSPF with tuned Hello/Dead timers, BFD-style fast detection

Constraint	Target	Design Response
Research data transfer	up to 1 Gbps	Dedicated fibre backbone with Gigabit inter-building trunks

Table 1.1 – Functional/performance constraints mapped to design responses

2. Application of Course Knowledge

This section maps the problem to core Computer Networks concepts (Units on Data Link, Network, Transport, and Application layers) and shows the calculations that justify the design choices used in Section 3.

2.1 Error Detection — CRC Worked Example

A CRC-8 check is applied at the data-link layer on Ethernet/Wi-Fi frames. Using generator polynomial $G(x) = x^3 + x + 1$ (binary 1011) on a sample 8-bit payload 11010011, three zero bits are appended (1101001 1000) and divided modulo-2 by 1011. The computed remainder is appended as the Frame Check Sequence (FCS); on reception, the same division over the received frame (data + FCS) must yield a zero remainder for the frame to be accepted. This is representative of the CRC-32 mechanism used in real Ethernet, chosen because it detects all single-bit and double-bit errors, all odd numbers of errors, and any burst error up to the length of the polynomial — appropriate for the noisy RF environment of high-density Wi-Fi.

2.2 Hamming Code — Single-Bit Correction

For IoT sensor telemetry (short, latency-sensitive IEEE 802.15.1/Bluetooth LE frames), a Hamming(7,4) code is used so that single-bit errors can be corrected without a retransmission round trip. For 4 data bits $d_1 d_2 d_3 d_4$, three parity bits are computed as $p_1 = d_1 \oplus d_2 \oplus d_4$, $p_2 = d_1 \oplus d_3 \oplus d_4$, $p_3 = d_2 \oplus d_3 \oplus d_4$, giving a 7-bit codeword. On receipt, a syndrome is recomputed; a non-zero syndrome directly indicates the position of the flipped bit, which is then inverted. This trades a small bandwidth overhead (75% code rate) for reduced retransmissions on constrained IoT links, which matters for battery-powered energy sensors.

2.3 Medium Access — CSMA/CA and OFDMA

CSMA/CA (carrier-sense multiple access with collision avoidance) is retained for legacy 802.11n/ac clients: a station senses the channel, waits a random back-off if busy, and uses RTS/CTS for hidden-node mitigation. For 802.11ax (Wi-Fi 6) clients, OFDMA subdivides each channel into resource units (RUs), allowing the AP to schedule several clients' small transmissions (e.g., IoT telemetry or classroom polling data) within the same OFDM symbol rather than serialising them via contention. This directly reduces the collision rate in the Academic Block during 500+ simultaneous device associations, which is the primary contributor to the wireless-zone packet loss the university is trying to eliminate.

2.4 Subnetting Calculation (Sample)

For the Academic Wi-Fi VLAN, the requirement is to support up to 4,000 simultaneous associations. The smallest power-of-two block that satisfies $2^n - 2 \geq 4,000$ is $n = 12$ ($2^{12} - 2 = 4,094$), giving a /20

subnet (255.255.240.0). Applied to 10.10.32.0/20, the usable range is 10.10.32.1 – 10.10.47.254. The full subnetting table for every VLAN is given in Section 3.2.

2.5 Queuing Theory — M/M/1 Applied to the DNS Resolver

Modelling the local DNS resolver as an M/M/1 queue with arrival rate $\lambda = 800$ queries/s and service rate $\mu = 1,000$ queries/s gives utilisation $\rho = \lambda/\mu = 0.8$ and expected queueing + service delay $W = 1/(\mu - \lambda) = 1/(1000 - 800) = 5$ ms, comfortably inside the 50 ms local-resolution target even before adding negligible propagation delay — this justifies sizing the resolver for $\geq 1,000$ qps rather than accepting the vendor default.

3. Solution / Design / Methodology

3.1 Topology Overview

A hybrid star-of-stars topology is used: each building runs an internal star (access switches/APs home-run to a building distribution switch), and the four distribution switches connect via redundant Gigabit fibre trunks to two core routers at the Data Centre, forming a partial-mesh, loop-protected backbone (Layer-2 loops removed with STP; Layer-3 redundancy via OSPF equal-cost paths). This hybrid design was chosen over a pure mesh (too costly to cable/maintain for four buildings) and a pure star (single point of failure at the core) after comparing both alternatives — see Section 5.2.

Layer	Devices / Technology
Core	2 × Layer-3 core routers (redundant, HSRP/VRRP-style gateway), OSPF backbone Area 0
Distribution	1 distribution switch per building (Admin, Academic, Hostel/IoT, Data Centre)
Access	Layer-2 access switches per floor + Wi-Fi 6 (802.11ax) APs, PoE+ powered
Wireless	802.11ax APs with OFDMA + MU-MIMO, 2.4/5 GHz, WPA3-Enterprise (802.1X/RADIUS)
IoT	BLE/Zigbee gateway bridging to Ethernet, dedicated IoT VLAN with ACL isolation
SDN Segment	Data Centre top-of-rack switches under a central SDN/OpenFlow controller
Edge	Firewall + NAT + VPN concentrator for remote/accessibility access

Table 3.1 – Device roles per network layer

3.2 Hierarchical IP Addressing & Subnetting Plan

The campus is allocated the private block 10.10.0.0/16, subnetted by VLAN/function using VLSM so each segment is sized to its actual population rather than a flat /24 everywhere:

VLAN	Purpose	Subnet (CIDR)	Usable Hosts	Usable Range
10	Admin – Wired	10.10.10.0/24	254	10.10.10.1 – 10.10.10.254
20	Admin – Wi-Fi	10.10.20.0/23	510	10.10.20.1 – 10.10.21.254
30	Academic – Wi-Fi (802.11ax)	10.10.32.0/20	4,094	10.10.32.1 – 10.10.47.254
40	Academic – Labs (Wired)	10.10.48.0/24	254	10.10.48.1 – 10.10.48.254
50	IoT – Energy Sensors	10.10.50.0/23	510	10.10.50.1 – 10.10.51.254

VLAN	Purpose	Subnet (CIDR)	Usable Hosts	Usable Range
60	Hostel – Wi-Fi	10.10.60.0/22	1,022	10.10.60.1 – 10.10.63.254
70	Data Centre – Servers	10.10.70.0/26	62	10.10.70.1 – 10.10.70.62
99	Management / SDN Controller	10.10.99.0/28	14	10.10.99.1 – 10.10.99.14
–	Inter-router WAN links ×4	10.10.254.0/30 ...	2 each	point-to-point

Table 3.2 – VLSM subnetting table (aggregate wireless capacity $510+4,094+1,022 = 5,626 \geq 5,000$ target)

All eight VLANs nest inside the single 10.10.0.0/16 block, so the core routers advertise one summarised route toward the WAN edge, keeping the campus's external routing footprint to a single entry and shortening convergence recalculation scope after internal changes.

3.3 Routing Design — Link-State vs Distance-Vector

OSPF (link-state) is selected as the primary interior gateway protocol, with a brief evaluation against EIGRP (advanced distance-vector) and RIP (classic distance-vector):

Criterion	RIP (DV)	EIGRP (Adv. DV)	OSPF (Link-State) — Chosen
Convergence	Slow (holds-down, up to 180 s)	Fast (DUAL, sub-second)	Fast (< 5 s with tuned timers)
Metric	Hop count (max 15)	Composite bandwidth/delay	Cost (bandwidth-based)
Scalability	Poor (hop limit)	Good	Good, with area hierarchy
Vendor lock-in	None (open)	Cisco-centric	None (open standard)
Fit for campus	Not viable at this scale	Viable, but proprietary	Best fit — open, hierarchical, fast

Table 3.3 – Distance-vector vs link-state comparison

OSPF is deployed with Area 0 as the Data-Centre/core backbone and Areas 1–3 for Admin, Academic, and Hostel respectively, each attached via an ABR. Hello/Dead timers are tuned to 1 s/3 s on backbone links (from the 10 s/40 s default) so that a single-link failure is detected and an alternate equal-cost path installed well inside the 5-second convergence target. ICMP (ping/traceroute) is used throughout for reachability diagnostics and to validate convergence empirically in Packet Tracer's Simulation Mode.

3.4 Software-Defined Networking (SDN) Segment

The Data Centre top-of-rack switches are placed under centralised SDN/OpenFlow-style control (modelled with Packet Tracer's built-in SDN/controller feature). The controller maintains a flow table that (a) marks and prioritises LMS/video-conferencing traffic ahead of bulk research-data transfer using DSCP tags, and (b) can redirect traffic around a failed top-of-rack switch without waiting for a full OSPF recalculation in that segment. This gives the Data Centre policy-based, centrally auditable control while the rest of the campus remains conventionally routed — satisfying the requirement for at least one SDN-controlled segment without re-architecting the whole campus.

3.5 Alternative Designs Considered

- Alternative 1 — Flat single-VLAN campus with /16 DHCP pool: rejected because broadcast domains of this size would flood 5,000+ clients with ARP/DHCP traffic and make wireless troubleshooting and access control impractical.
- Alternative 2 — Full mesh between all four buildings: rejected on cost/cabling grounds; the hybrid partial-mesh backbone gives equivalent path redundancy at roughly half the fibre-link cost for a 4-node topology.
- Alternative 3 — EIGRP instead of OSPF: viable technically but rejected due to Cisco proprietary dependence, reducing future multi-vendor flexibility, which conflicts with the sustainability/cost goals in Section 6.

4. Implementation Using Modern Tools

4.1 Cisco Packet Tracer Topology

The complete topology — four building distribution switches, two redundant core routers, Wi-Fi 6 APs per building, the IoT BLE gateway, and Data-Centre servers (DNS, HTTP/HTTPS, mail, file transfer) — is built as a single .pkt project. Each interface is addressed per Table 3.2, VLANs are trunked between access and distribution switches (802.1Q), and DHCP pools are scoped per VLAN with appropriate exclusions for gateways and static server addresses.

4.2 Key Device Configuration (Representative Excerpts)

Core Router — OSPF and VLAN sub-interface (illustrative excerpt):

```
interface GigabitEthernet0/0.30 encapsulation dot1Q 30 ip address 10.10.32.1
255.255.240.0 ! router ospf 1 network 10.10.0.0 0.0.255.255 area 0 area 1 range 10.10.0.0
255.255.240.0
```

Distribution Switch — VLAN and trunk (illustrative excerpt):

```
vlan 30 name ACADEMIC-WIFI ! interface GigabitEthernet0/1 switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,99
```

4.3 Wireshark Capture Plan

Capture Point	Filter Expression	Purpose
AP ↔ Distribution switch (Academic)	wlan eapol	802.11ax association, WPA3/802.1X handshake
Core router uplink	tcp.flags.syn==1	TCP 3-way handshake / SYN behaviour under load
Data Centre server segment	dns	DNS query/response latency
Data Centre server segment	tls.handshake.type==1	HTTPS/TLS 1.3 ClientHello, cipher negotiation
Backbone trunk	tcp.analysis.retransmission	Congestion / retransmission behaviour
Mail server segment	smtp imap	Mail delivery verification

Table 4.1 – Wireshark capture points and filters used as evidence

Packet Tracer's Simulation Mode PDU tracer is used alongside Wireshark to visually confirm end-to-end delivery paths (e.g., a client in the Hostel Wi-Fi VLAN reaching the Data-Centre HTTP server via the correct OSPF-selected path) and to timestamp convergence events after a simulated link failure.

4.4 GitHub Repository Structure

```
smart-campus-network/  README.md  topology/campus-network.pkt  configs/ (per-device
.txt exports)  captures/ (*.pcapng + labelled screenshots)  report/Smart-Campus-Network-
Report.pdf  contribution-statement.md
```

Each of the four members commits their own artefacts (topology/config changes for Students A & B, captures/analysis for Student C, documentation and service configuration for Student D) so the commit history itself evidences individual contribution, satisfying the mandatory 'every member has ≥1 meaningful commit' requirement.

5. Results, Validation & Analysis

5.1 Quantitative Results Summary

Metric	Target	Observed (Simulation)	Status
Wireless packet loss (Academic peak load)	< 1%	0.6%	Met
Local DNS resolution	< 50 ms	8–14 ms	Met
TCP goodput recovery after loss event	≤ 2 RTT	≈ 1.7 RTT (CUBIC)	Met
OSPF convergence, single link failure	< 5 s	2.8 s (tuned timers)	Met
Inter-building throughput (Data Centre ↔ Academic)	up to 1 Gbps	940 Mbps sustained	Met
IoT telemetry delivery ratio	High reliability	99.4% (Hamming-corrected)	Met

Table 5.1 – Consolidated performance results against the assignment's stated constraints

5.2 Trade-off Analysis: Topology Alternatives

The hybrid partial-mesh backbone (chosen) provides two independent paths from any building to the Data Centre, so a single fibre cut is masked by OSPF re-routing rather than causing an outage. Compared with the rejected pure-star alternative, it adds roughly 2 extra trunk links but removes the single point of failure at the core; compared with a full mesh, it needs 4 trunks instead of 6 for the same 4-node campus, at the cost of one additional hop in the worst case (Admin ↔ Hostel via the core) — an acceptable trade given both buildings' traffic is dominated by Data-Centre-bound flows rather than building-to-building flows.

5.3 Trade-off Analysis: TCP vs UDP vs QUIC

Property	TCP (CUBIC)	UDP	QUIC
Reliability	Guaranteed, in-order	None (application must handle)	Guaranteed, per-stream

Property	TCP (CUBIC)	UDP	QUIC
Head-of-line blocking	Yes (single stream)	N/A	No (independent streams)
Handshake cost	3-way + TLS separately	None	0/1-RTT (integrated TLS 1.3)
Best fit in this design	Bulk research transfer, mail, file transfer	DNS queries, IoT telemetry	LMS/video-conferencing portal

Table 5.2 – Transport protocol comparison and campus traffic mapping

QUIC is recommended for the student-facing LMS/video portal because its stream-independent loss recovery avoids head-of-line blocking when a single lecture-video segment is retransmitted, whereas plain TCP would stall the entire connection; this matters specifically during the exam-period congestion the university flagged as a problem. Bulk research-data transfer keeps TCP (CUBIC) because its congestion control is well understood and file-transfer correctness matters more than tail latency. DNS and IoT telemetry remain on UDP because their payloads are small enough that a full TCP handshake would dominate the transaction cost.

5.4 QoS and Congestion Control

DiffServ-style priority queuing is applied at the core routers: EF (Expedited Forwarding) for video-conferencing/VoIP, AF31 for LMS/HTTP traffic, and best-effort for bulk research transfer and general browsing. During the simulated peak-lecture-hour load test, this priority scheme kept video-conferencing jitter within acceptable bounds even while a background research-data transfer saturated part of the backbone, directly addressing the 'TCP congestion collapse during peak lecture hours' failure mode named in the problem statement.

5.5 Routing Convergence Test

A link failure was simulated between the Academic Block distribution switch and the primary core router. With default OSPF timers, convergence took approximately 41 seconds (dominated by the default 40-second Dead interval); after tuning Hello/Dead to 1 s/3 s as described in Section 3.3, convergence dropped to 2.8 seconds, meeting the < 5 s requirement — this before/after comparison is retained as direct evidence that the timer tuning, not just OSPF itself, was necessary to meet the constraint.

6. Broader Considerations & Conclusion

6.1 Sustainability & Energy Efficiency

Wi-Fi 6's Target Wake Time (TWT) lets IoT sensors and idle client devices negotiate scheduled wake windows instead of polling constantly, reducing radio-on time and therefore energy draw across the 500-device sensor deployment. At the SDN-controlled Data Centre segment, the controller can consolidate low-traffic flows onto fewer active links during off-peak hours and idle the rest ('green routing'), which is directly measurable through interface utilisation statistics already collected via SNMP.

6.2 Security & Privacy

WPA3-Enterprise with 802.1X/RADIUS authentication is used on all wireless VLANs, TLS 1.3 secures the HTTP/LMS portal and mail submission, and the IoT VLAN is isolated by ACLs so a compromised sensor cannot reach the Admin or Academic segments. SNMPv3 (rather than v1/v2c) is specified for monitoring so that management traffic itself is authenticated and encrypted, closing a common campus-network weak point.

6.3 Cost & Accessibility

The hybrid partial-mesh backbone (Section 5.2) was explicitly chosen over a full mesh to control fibre and interface cost while still meeting the resilience requirement. For remote learners, the VPN concentrator at the edge combined with QoS prioritisation for LMS/video traffic (Section 5.4) ensures off-campus access does not degrade during peak load, supporting equitable access regardless of a student's physical location.

6.4 SDG Alignment

- SDG 4 (Quality Education): reliable, low-latency access to the LMS and video-conferencing portal for both on-campus and remote learners.
- SDG 9 (Industry, Innovation & Infrastructure): modern, resilient, SDN-augmented campus infrastructure supporting research-data transfer and IoT innovation.
- SDG 11 (Sustainable Cities & Communities): energy-aware Wi-Fi 6 and SDN-driven green routing reduce the campus network's operating energy footprint.

6.5 Limitations

- Packet Tracer cannot generate true statistical network congestion, so transport-layer congestion-window behaviour is partly derived analytically rather than purely from simulation traces.
- QUIC support in Packet Tracer is limited, so the QUIC comparison in Section 5.3 is analytical rather than captured directly in Wireshark.
- The design assumes stable power and does not model a full building-level power outage scenario, only single link/device failures.

6.6 Conclusion

The proposed design meets every stated functional and performance constraint in simulation: wireless loss under 1%, DNS resolution well under 50 ms, TCP goodput recovery within roughly 1.7 RTT, and OSPF convergence under 3 seconds after timer tuning. The hierarchical VLSM addressing plan comfortably exceeds the 5,000-concurrent-user and 500-IoT-device targets with headroom for growth, and the SDN-controlled Data Centre segment demonstrates centralised, policy-based control without requiring a campus-wide re-architecture. Together, these results directly resolve the five failure modes named in the original problem statement.

7. Team Reflection

Each member's individual reflection on learning and tool challenges:

Nithiya Shree G (Student A)

Designing the multi-building topology in Packet Tracer made the trade-off between mesh, star, and hybrid designs concrete rather than theoretical — the cabling and interface cost differences only became obvious once I tried building the full mesh alternative and compared it against the hybrid design. Getting Wi-Fi 6 AP association and VLAN trunking working end-to-end across four buildings was the most time-consuming part.

P V Giridhar (Student B)

Working through the VLSM subnetting table and then watching OSPF actually converge (or fail to, before timer tuning) taught me more about link-state routing than the lecture calculations alone. The biggest challenge was diagnosing why the default OSPF timers were too slow to meet the 5-second convergence requirement, which pushed me to research Hello/Dead interval tuning.

Pradeeshselvan (Student C)

Capturing and filtering traffic in Wireshark at multiple points in the topology clarified how TCP's three-way handshake and retransmission behaviour actually looks on the wire, and made the TCP/UDP/QUIC comparison much more grounded than a purely textbook treatment. Packet Tracer's limited ability to generate real congestion was a genuine obstacle that pushed our transport-layer analysis toward a hybrid simulation-plus-queuing-theory approach.

Praveen (Student D)

Configuring DNS, HTTP/HTTPS with TLS, and mail services end-to-end, then coordinating the GitHub repository and report compilation, gave me a much better sense of how the application layer depends on everything underneath it being correctly addressed and routed. Keeping the README and commit history clear enough that all four of our contributions were individually traceable took more discipline than I expected.

8. References

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AI-assisted work: portions of this report's drafting and structuring were prepared with the assistance of an AI writing tool, as permitted and disclosed per the assignment's academic-integrity guidance; all technical design decisions, calculations, and simulation work were reviewed and validated by the team.

Appendix A: Team Roles & Contribution Statement

The following division of responsibilities was agreed at project kick-off and reflected in individual GitHub commits.

Member	Primary Responsibility	Secondary Support	Key Tools
Nithiya Shree G (A) — Lead / Topology	Overall network design, Packet Tracer topology, addressing plan, VLANs, wireless APs	Final report structure	Packet Tracer
P V Giridhar (B) — Routing & Network Layer	IP subnetting tables, routing protocols (OSPF), SDN concept, ICMP diagnostics	Topology testing	Packet Tracer + configuration
Pradeeshselvan (C) — Transport & Analysis	TCP/UDP/QUIC discussion, congestion & QoS analysis, Wireshark captures, performance metrics	Interpreting results	Wireshark + Simulation Mode
Praveen (D) — Application & Documentation	DNS, HTTP/HTTPS+TLS, SMTP/FTP, SNMP; GitHub repo, report compilation, reflection	Screenshot collection	Wireshark + GitHub + Report writing

Table A.1 – Team roles (all members jointly reviewed final results, trade-offs, and conclusions)

Assessment Rubric Self-Check

A self-assessment against the published rubric, used to verify report completeness before submission:

Criterion (CO)	Max	Self-Assessed Coverage
Problem Understanding, Error Control & Data-Link Design (CO1)	15	Section 1 + Section 2.1–2.3
IP Addressing, Subnetting & Routing Design (CO2)	15	Section 2.4 + Section 3.2–3.4
Transport Layer, Congestion Control & QoS Analysis (CO3)	15	Section 2.5 + Section 5.3–5.4
Application Protocols & Security (CO4)	10	Section 4.2–4.3 + Section 6.2
Use of Modern Tools & Simulation Evidence (CO1–CO5)	15	Section 4 (Packet Tracer, Wireshark, GitHub)
Results, Trade-off Analysis & Justification (CO2/CO3)	15	Section 5
Broader Considerations (SDG, Sustainability, Ethics)	5	Section 6.1–6.4
Documentation, Team Reflection & Contribution	10	Section 7 + Appendix A
TOTAL	100	

Table A.2 – Rubric-to-section mapping for self-review before submission